ASSIGNMENT -9

Formation of Matrices and vectors using MATLAB and carrying out operations involving matrices.

1. Create a 10x10 random matrix and do the following:
   i. Multiply all the elements by 100 and then round off all elements of the matrix to integers.
   ii. Replace all the elements of the matrix less than 10 to zeros.
   iii. Extract all the elements of the matrix that are between 30 to 50 and put them in a vector
2. Create a matrices A of size 12 × 10 & display it in command window.
3. Display the value of A(8,9), A(3,6), A(10,11), A(8,7), A(9,12) of the above matrix.
4. Make the above positions in A to 0.
5. Display sub-matrix of A for the range A(2:3,1:5).
6. Display the whole matrix starting from the element A(2,3).
7. Delete a specified row & column in the above matrix.
8. Create matrix using the following commands
   zeros, eyes, ones & write the purpose of the above commands.
9. Create an Identity Matrix.
10. Create diagonal matrix.
11. Create a matrix of all elements same.
12. Create a vector using command linspace & logspace. Write the syntax and purpose of these commands.
13. Check the following arithmetic expression for matrix.(True/False)
    i. A + B = = B + A
    ii. A - B = = B - A
    iii. A + (B + C) == (A + B) + C
    iv. A * (B + C) == A*B + A*C
    v. (B + C) / A == B/A + C/A
    vi. A*B == B*A
    vii. A*1 == A
    viii. $A*A^{-1}$ == 1
    ix. A == A^T
    x. A == $-A^T$

14. Create a 10x10 random matrix and do the following:
    i. Multiply all the elements by 100 and then round off all elements of the matrix to integers.
    ii. Replace all the elements of the matrix less than to zeros.
    iii. Extract all the elements of the matrix that are between 30 to 50 and put them in a vector

15. A = [1 2; 3 4; 5 6], B = [1 2 3; 4 5 6]
    i. Compute $(AB)^{20}$
    ii. Find the eigen values for $(BA)^{6}$ and for each eigen value a corresponding eigen vector.

16. Create a 20 X 20 matrix with the command A= ones(20). Now replace the 10 X 10 sub-matrix between rows 6:15 and columns 6:15 with zeros.

    See the structure of the matrix in terms of non-zero entries.

    Set the 5 X 5 sub-matrices in the top right corner and bottom left corner to zeros.